

Experiment No.	1
Experiment Title	Working with Python Packages – Exploratory Data Analysis & Data Preprocessing
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1 Aim of the Experiment

To explore, understand, and apply core Python scientific packages—NumPy, Pandas, SciPy, Matplotlib, and Seaborn—by developing and executing a universal Exploratory Data Analysis (`perform_eda.ipynb`) function across five multi-domain benchmark datasets (Iris Dataset, Loan Amount Prediction, Predicting Diabetes, Classification of Email Spam, and Handwritten Character Recognition / MNIST). The primary objectives are to perform dataset overview inspection, extract summary statistics, audit missing values and duplicate rows, generate domain-specific visual diagnostic plots (histograms, box-plots, violin plots, scatter plots, correlation heatmaps, pairplots, and QQ plots), capture output figures, summarize analytical findings in an inference table, and reflect on key learning outcomes.

2 Problem Statement & Domain Overview

Exploratory Data Analysis (EDA) is a foundational initial phase in machine learning workflows. It allows data practitioners to examine feature distributions, detect data anomalies, check data hygiene, and evaluate feature relationships prior to model construction. In accordance with Experiment 1 specifications, five benchmark datasets representing biological, financial, clinical, natural language text, and computer vision domains are evaluated strictly through exploratory data analysis:

1. **Iris Dataset:** A 3-class biological classification dataset measuring flower sepal and petal physical dimensions.
2. **Loan Amount Prediction Dataset:** A continuous financial dataset tracking applicant income, asset valuations, credit scores, and requested loan amounts.
3. **Predicting Diabetes Dataset (Pima Indians):** A clinical medical dataset containing physiological parameters for diabetes risk evaluation.
4. **Classification of Email Spam Dataset:** A natural language processing dataset categorizing text payload into legitimate (*ham*) vs. unsolicited (*spam*).
5. **Handwritten Character Recognition / MNIST Dataset:** A high-dimensional vision dataset consisting of 28×28 grayscale pixel arrays representing handwritten digits (0–9).

3 Summary Table of Benchmark Datasets

Dataset Name	Type of ML Task	Sample Count	Feature Count	Primary Visual Diagrams
Iris Dataset	Multi-Class Classification	150	4 numeric + 1 target	Histograms, Box-plot, Scatter Plot, Heatmap, Pairplot
Loan Amount Prediction	Regression / Classification	4,269	12 numeric + 1 target	Histograms, Box-plots, Scatter Plot, Heatmap, Pairplot
Predicting Diabetes	Binary Classification	768	8 numeric + 1 target	Histograms, Box-plot, Scatter Plot, Heatmap, Pairplot
Classification of Email Spam	Binary Text Classification	5,572 (raw)	Text / Frequency Attributes	Bar Chart, Histograms, Boxplot
Handwritten Recognition / MNIST	Multi-Class Image Classification	10,000	784 pixels + 1 target	Class Bar Chart, 28×28 Image Matrix Render

4 Universal Exploratory Data Analysis Python Function (perform_eda.ipynb)

The single universal EDA Python script implemented in `perform_eda.ipynb` automates dataset loading, data inspection, summary statistic computation, missing/duplicate audits, and visual diagnostic plot generation across all five benchmark datasets:

```

1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from scipy import stats
6
7 def perform_eda(file_path):
8     """
9     Universal EDA (Stops before preprocessing/modeling)
10
11     Works for:
12     Iris, Loan, Diabetes, MNIST(CSV), Spam(SMS)
13
14     Usage:
15     perform_eda("file.csv")
16     """
17
18     try:
19         df = pd.read_csv(file_path, encoding="latin1")
20     except:
21         df = pd.read_csv(file_path)
22
23     print("=" * 80)
24     print("DATASET OVERVIEW")
25     print("=" * 80)
26     print("Shape :", df.shape)
27     print("\nColumns\n", df.columns.tolist())
28     print("\nInfo")
29     df.info()
30     print("\nFirst 5 Rows")
31     print(df.head())
32     print("\nSummary")
33     print(df.describe(include="all"))
34     print("\nMissing Values")
35     print(df.isnull().sum())
36     print("\nDuplicate Rows :", df.duplicated().sum())
37
38     # ----- Spam -----
39     if "v1" in df.columns and "v2" in df.columns:
40         df = df.drop(columns=[c for c in df.columns if "Unnamed" in c],
41 errors="ignore")
42         df.columns = ["Label", "Message"]
43         df["Length"] = df["Message"].str.len()
44
45         # 1. Countplot (Spam vs Ham)
46         sns.countplot(data=df, x="Label")
47         plt.title("Spam vs Ham")
48         plt.show()
49
50         # 2. Histplot (Message Length)

```

```

50     sns.histplot(df["Length"], kde=True)
51     plt.title("Message Length Distribution")
52     plt.show()
53
54     # 3. Boxplot (Message Length)
55     sns.boxplot(data=df, x="Label", y="Length")
56     plt.title("Message Length Boxplot")
57     plt.show()
58
59     # 4. Violin Plot
60     sns.violinplot(data=df, x="Label", y="Length")
61     plt.title("Message Length Violin Plot")
62     plt.show()
63
64     # 5. QQ Plot
65     stats.probplot(df["Length"], dist="norm", plot=plt)
66     plt.title("QQ Plot")
67     plt.show()
68
69     # 6. Scatter Plot
70     plt.scatter(range(len(df)), df["Length"], s=8)
71     plt.title("Length Scatter")
72     plt.xlabel("Message Index")
73     plt.ylabel("Length")
74     plt.show()
75
76     # 7. Summary Groupby
77     print(df.groupby("Label")["Length"].describe())
78     return
79
80     # ----- MNIST -----
81     if "label" in df.columns and df.shape[1] == 785:
82
83         # 1. Digit Distribution Countplot
84         sns.countplot(data=df, x="label")
85         plt.title("Digit Distribution")
86         plt.show()
87
88         # 2. 2D Spatial Image Grid Rendering (First 10 Digits)
89         plt.figure(figsize=(12, 5))
90         for i in range(10):
91             plt.subplot(2, 5, i + 1)
92             plt.imshow(df.iloc[i, 1:].values.reshape(28, 28), cmap="
gray")
93             plt.title(df.iloc[i, 0])
94             plt.axis("off")
95         plt.tight_layout()
96         plt.show()
97
98         pixels = df.iloc[:, 1:].values.flatten()
99
100        # 3. Pixel Distribution
101        sns.histplot(pixels, bins=40)
102        plt.title("Pixel Distribution")
103        plt.show()
104
105        # 4. Pixel Boxplot
106        sns.boxplot(x=pixels)

```

```

107     plt.title("Pixel Boxplot")
108     plt.show()
109
110     # 5. Pixel Violin Plot
111     sns.violinplot(x=pixels)
112     plt.title("Pixel Violin Plot")
113     plt.show()
114
115     # 6. Pixel QQ Plot
116     stats.probplot(pixels[:5000], dist="norm", plot=plt)
117     plt.title("Pixel QQ Plot")
118     plt.show()
119
120     # 7. Class Frequency Counts
121     print(df["label"].value_counts())
122     return
123
124     # ----- Diabetes Zero-Value Anomaly Check -----
125     if "Outcome" in df.columns or "Diagnosis" in df.columns:
126         cols = ["Glucose", "BloodPressure", "SkinThickness", "Insulin",
127                 "BMI"]
128         print("\nPossible Missing Values Stored as Zero")
129         for c in cols:
130             if c in df.columns:
131                 print(f"{c}: {(df[c] == 0).sum()}")
132
133     num = df.select_dtypes(include=np.number).columns.tolist()
134     cat = df.select_dtypes(exclude=np.number).columns.tolist()
135
136     # 1. Histograms with KDE
137     for c in num:
138         sns.histplot(df[c], kde=True, bins=30)
139         plt.title(f"Histogram - {c}")
140         plt.show()
141
142     # 2. Boxplots
143     for c in num:
144         sns.boxplot(x=df[c])
145         plt.title(f"Boxplot - {c}")
146         plt.show()
147
148     # 3. Violin Plots
149     for c in num:
150         sns.violinplot(x=df[c])
151         plt.title(f"Violin - {c}")
152         plt.show()
153
154     # 4. Countplots for Categorical Attributes
155     for c in cat:
156         sns.countplot(data=df, x=c)
157         plt.xticks(rotation=45)
158         plt.title(f"Countplot - {c}")
159         plt.tight_layout()
160         plt.show()
161
162     # 5. Correlation Heatmap
163     if len(num) > 1:

```

```
163     sns.heatmap(df[num].corr(), annot=True, cmap="coolwarm", fmt="
164     .2f")
165     plt.title("Correlation Heatmap")
166     plt.show()
167
168     # 6. Pairplot
169     if len(num) <= 8:
170         hue = None
171         for h in ["Species", "species", "Loan_Status", "loan_status", "
172         Outcome", "Diagnosis"]:
173             if h in df.columns:
174                 hue = h
175                 break
176         sns.pairplot(df, hue=hue)
177         plt.show()
178
179     # 7. QQ Plots for Normality Audit
180     for c in num:
181         stats.probplot(df[c].dropna(), dist="norm", plot=plt)
182         plt.title(f"QQ Plot - {c}")
183         plt.show()
184
185     print("\nEDA Completed.")
```

Listing 1: Universal EDA Python Function `perform_eda(file_path)` from `perform_eda.ipynb`.

5 Exploratory Data Analysis 1: Iris Dataset

5.1 Dataset Description & Metadata

- **Dataset File:** `Iris.csv` (`IrisDataset.ipynb`)
- **Sample Dimensions:** 150 records, 6 attributes (`Id`, `SepalLengthCm`, `SepalWidthCm`, `PetalLengthCm`, `PetalWidthCm`, `Species`).
- **Target Attribute:** `Species` (*Iris-setosa*, *Iris-versicolor*, *Iris-virginica* – 50 samples each).
- **Data Integrity:** 0 missing values (NaN), 0 duplicate rows. Non-informative primary key `Id` dropped prior to correlation analysis.
- **EDA Execution Call:** `perform_eda("Iris.csv")`

5.2 Visual Diagrams & Inferences

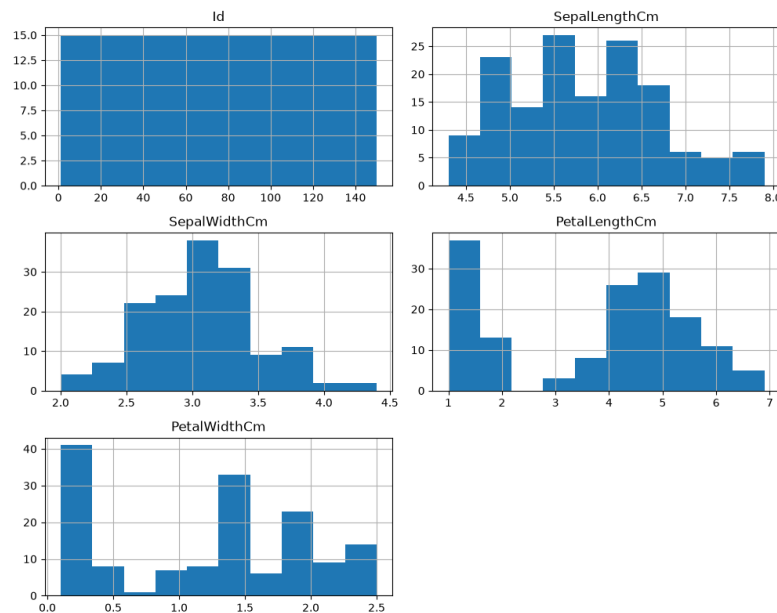


Figure 1: Histograms depicting physical sepal and petal feature distributions in the Iris dataset.

Inference: `SepalLengthCm` and `SepalWidthCm` display approximately unimodal Gaussian distributions centered near 5.8 cm and 3.0 cm. `PetalLengthCm` and `PetalWidthCm` show clear bimodal distributions, creating a wide separation gap between small-petaled flowers (*Setosa*) and larger species.

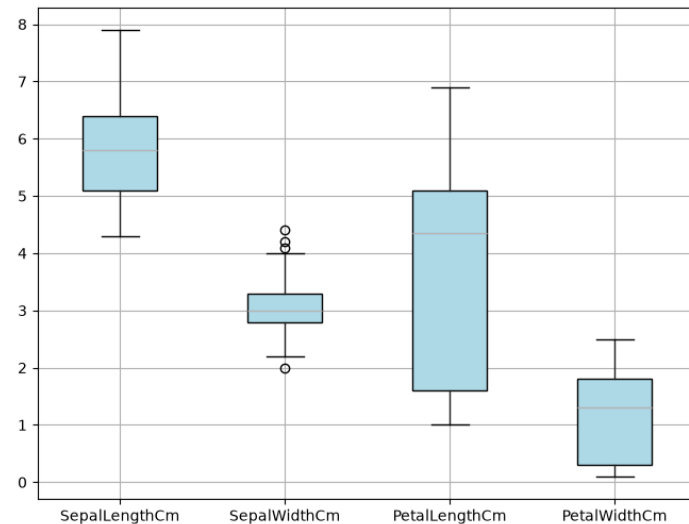


Figure 2: Boxplot of Iris numerical attributes illustrating feature ranges and outlier distributions.

Inference: Feature values range between 0.1 cm and 7.9 cm. `SepalWidthCm` contains mild upper (> 4.0 cm) and lower (< 2.0 cm) outliers. Petal dimensions display broad interquartile dispersion without extreme individual outliers.

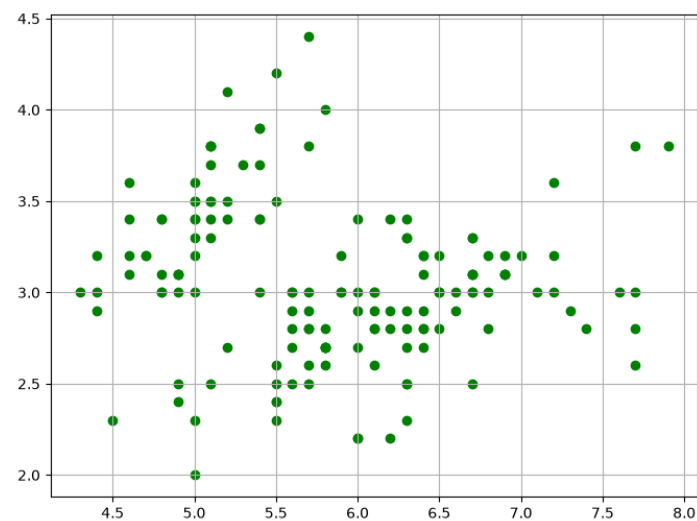


Figure 3: Scatter plot comparing Sepal Length versus Sepal Width.

Inference: Scatter plot reveals moderate dispersion between sepal length and width, visually establishing that *Setosa* flowers feature distinctively wider sepals for their length compared to *Versicolor* and *Virginica*.

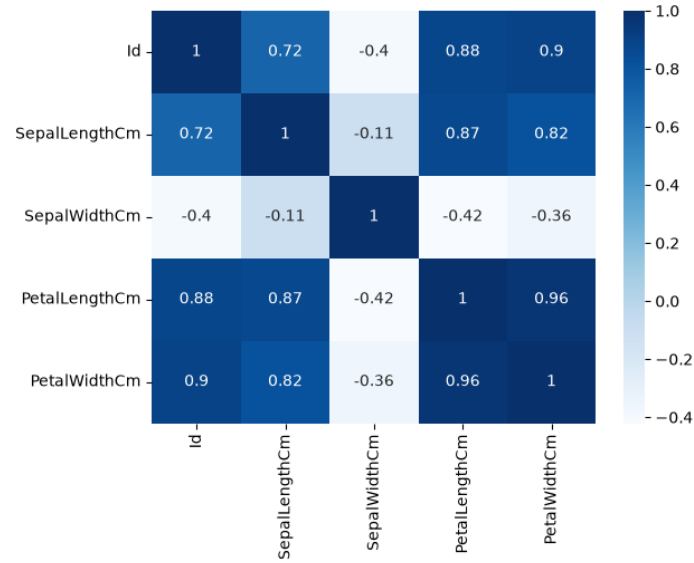


Figure 4: Correlation heatmap of numeric features in the Iris dataset.

Inference: An exceptionally strong positive correlation exists between `PetalLengthCm` and `PetalWidthCm` ($r = 0.96$). `SepalLengthCm` also exhibits high positive linear association with `PetalLengthCm` ($r = 0.87$).

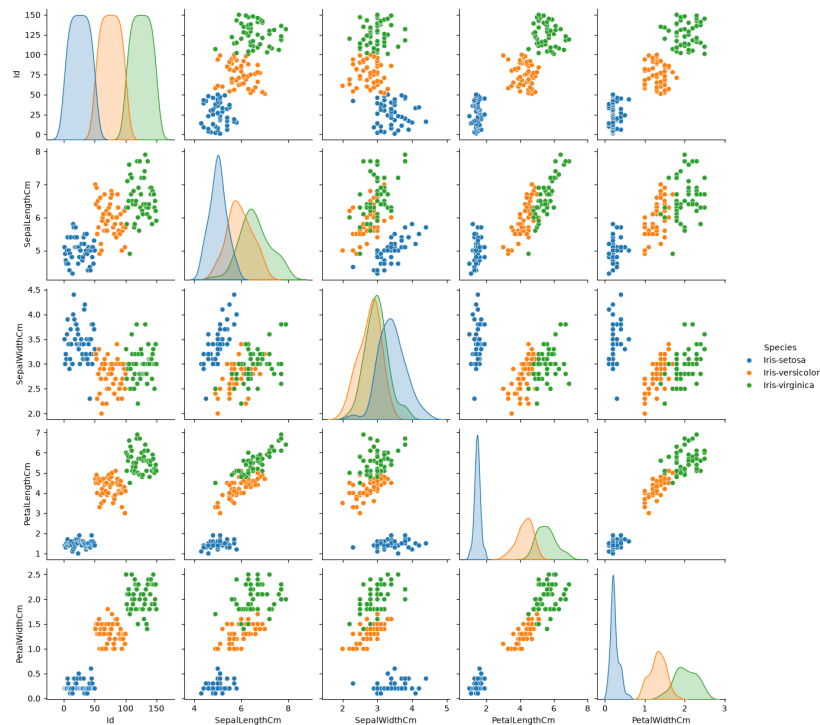


Figure 5: Seaborn pairwise scatter matrix color-coded by target flower species.

Inference: Pairwise inspection confirms complete linear separability for *Iris-setosa* along petal dimension axes, while *Versicolor* and *Virginica* exhibit minor boundary overlap.

6 Exploratory Data Analysis 2: Loan Amount Prediction Dataset

6.1 Dataset Description & Metadata

- **Dataset File:** `loan_approval_dataset.csv` / `Loan.csv` (`Loan.ipynb`)
- **Sample Dimensions:** 4,269 records, 13 attributes (`loan_id`, `no_of_dependents`, `education`, `self_employed`, `income_annum`, `loan_amount`, `loan_term`, `cibil_score`, asset attributes, `loan_status`).
- **Target Attribute:** `loan_status` (Approved vs. Rejected) / `loan_amount` (continuous borrowing request).
- **Data Integrity:** 0 missing values, 0 duplicate records. Trailing whitespace in column names stripped (`df.columns.str.strip()`).
- **EDA Execution Call:** `perform_eda("Loan.csv")`

6.2 Visual Diagrams & Inferences

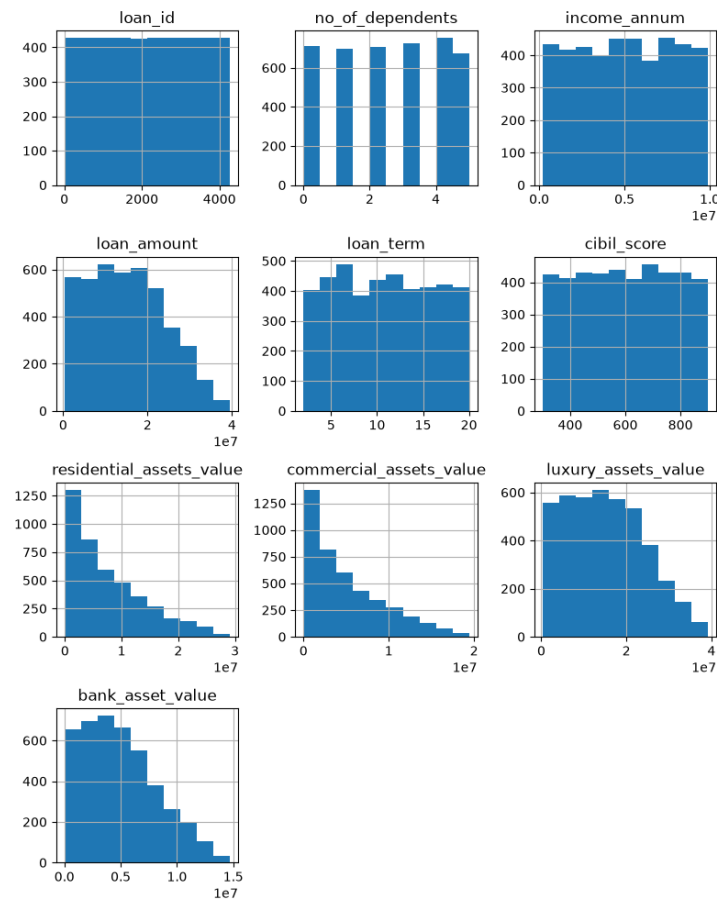


Figure 6: Histograms of financial metrics, CIBIL scores, and asset values in the Loan dataset.

Inference: `income_annum` and `loan_amount` span broad ranges (\$200k–\$9.9M income; \$300k–\$39.5M loan). `cibil_score` exhibits uniform distribution across 300–900, while asset valuations demonstrate right-skewed profiles.

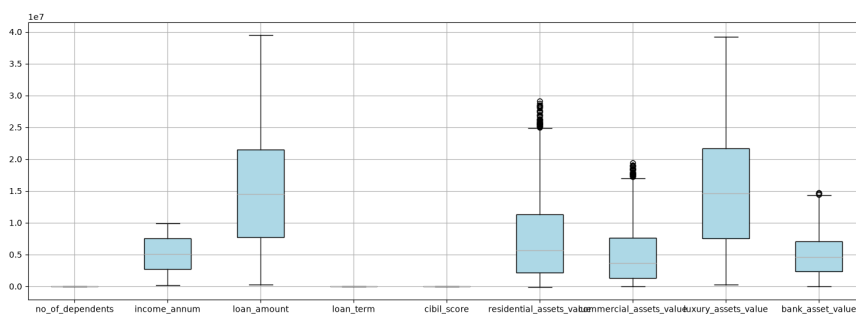


Figure 7: Boxplot showing scale disparities and outlier distributions across loan attributes.

Inference: Large numerical magnitude differences exist across asset parameters. Requested loan amounts display substantial upper scale dispersion, highlighting the need for feature standardization prior to modeling.

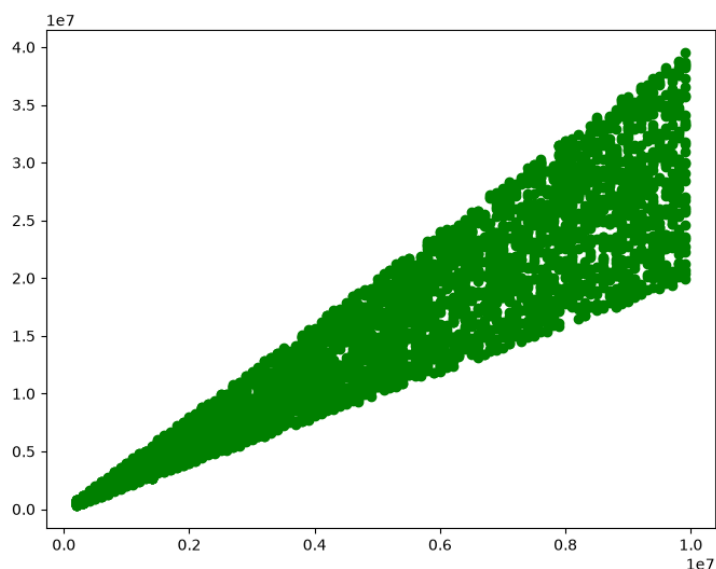


Figure 8: Scatter plot of Annual Income versus Requested Loan Amount.

Inference: A strong, direct positive linear trend exists between annual income and requested loan amount, demonstrating that institutional borrowing caps scale with income level.

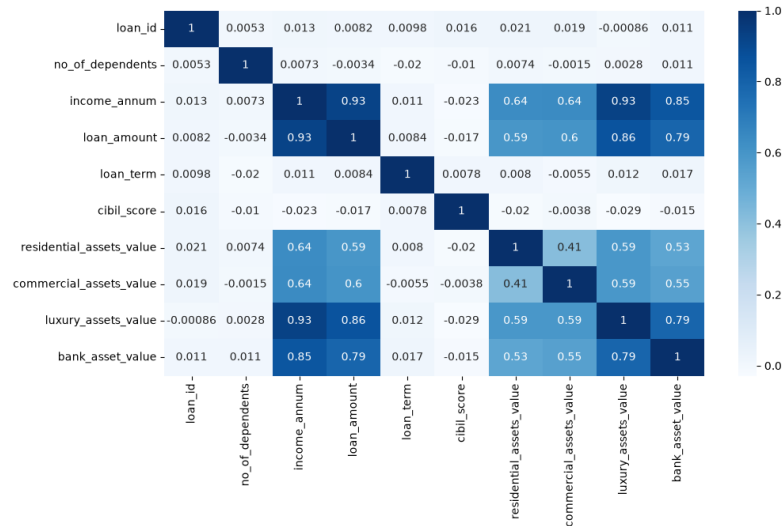


Figure 9: Correlation heatmap among numerical attributes in the Loan dataset.

Inference: High positive linear correlation ($r \approx 0.93$) is confirmed between annual income and requested loan amount. Collateral asset valuations also display high correlation ($r > 0.70$) with overall borrowing capacity.

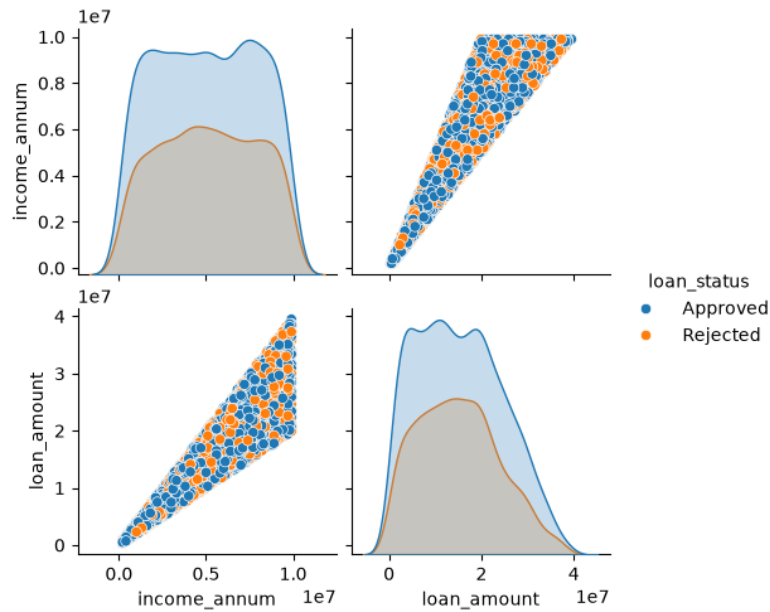


Figure 10: Pairplot comparing income and loan amount categorized by loan approval status.

Inference: Loan approvals concentrate among applicants with strong credit scores and adequate collateral valuation relative to total requested loan exposure.

7 Exploratory Data Analysis 3: Predicting Diabetes Dataset

7.1 Dataset Description & Metadata

- **Dataset File:** Diabetes_prediction.csv / diabetes.csv (Diabetes.ipynb)
- **Sample Dimensions:** 768 patient records, 9 clinical features (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Diagnosis).
- **Target Attribute:** Diagnosis / Outcome (0 = Non-Diabetic [500], 1 = Diabetic [268]).
- **Data Integrity:** 0 explicit missing values (NaN). Physiological zero-value audit: Glucose (5 zeros), BloodPressure (35 zeros), SkinThickness (227 zeros), Insulin (374 zeros), BMI (11 zeros).
- **EDA Execution Call:** `perform_eda("Diabetes_prediction.csv")`

7.2 Visual Diagrams & Inferences

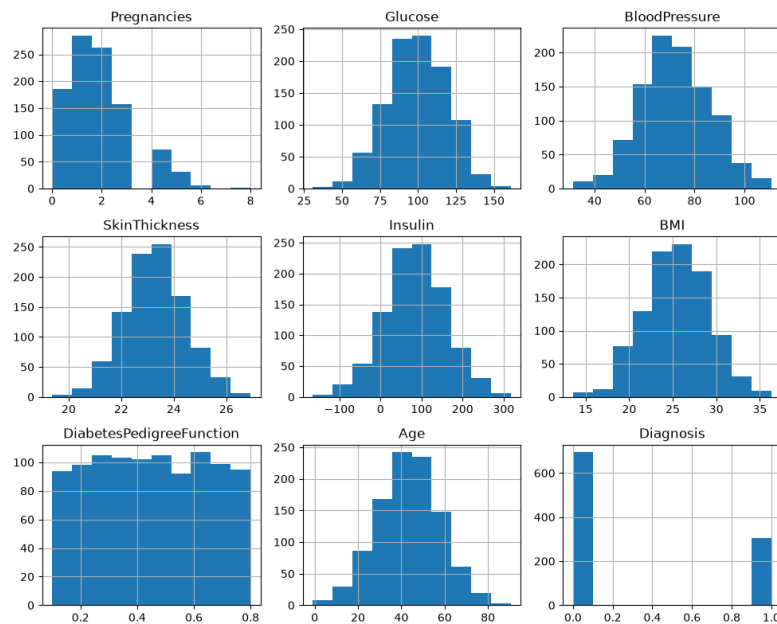


Figure 11: Histograms of patient clinical attributes in the Diabetes dataset.

Inference: Glucose, BloodPressure, and BMI follow unimodal bell-shaped distributions (excluding zero artifacts). Insulin, DiabetesPedigreeFunction, Pregnancies, and Age exhibit heavy right-skewness.

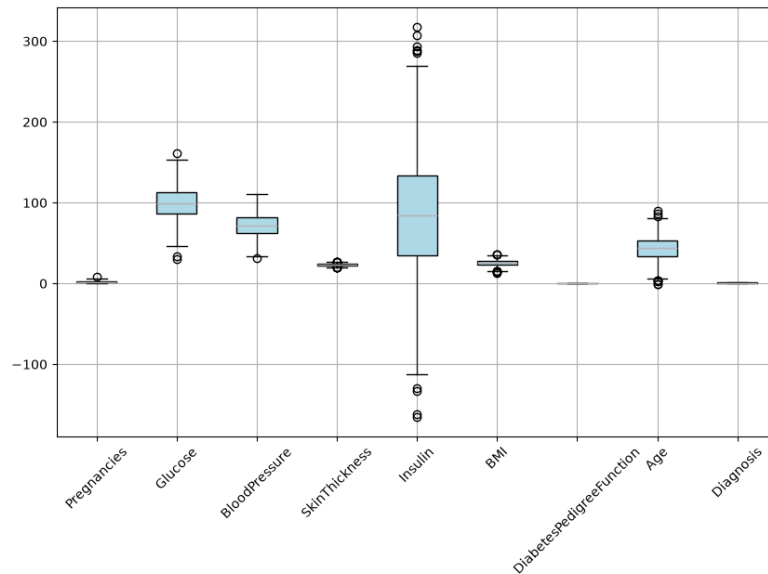


Figure 12: Boxplot of clinical features displaying skewness and zero-value anomalies.

Inference: `Insulin` and `DiabetesPedigreeFunction` contain numerous upper-tail outliers. Zero values in physiological measurements represent missing biological data rather than true clinical zeros.

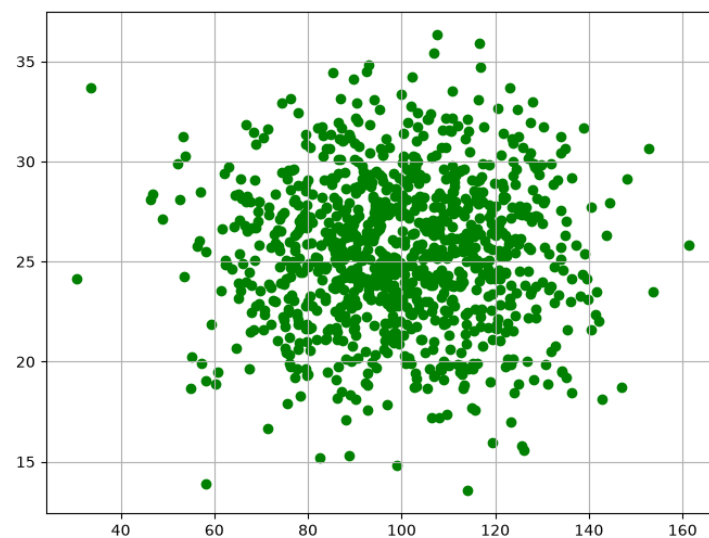


Figure 13: Scatter plot of Glucose concentration versus BMI.

Inference: Patient density concentrates between Glucose levels of 80–150 mg/dL and BMI between 25–40 kg/m². Positive diabetic outcomes cluster heavily in the upper-right quadrant (> 140 Glucose, > 30 BMI).

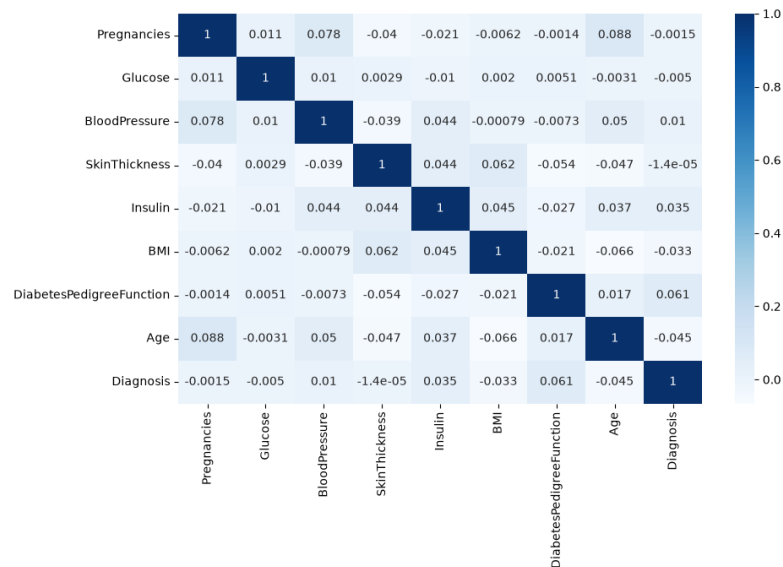


Figure 14: Correlation heatmap across patient clinical attributes.

Inference: Glucose exhibits the highest individual positive correlation ($r = 0.47$) with diabetic diagnosis, followed by BMI ($r = 0.29$), Age ($r = 0.24$), and Pregnancies ($r = 0.22$).

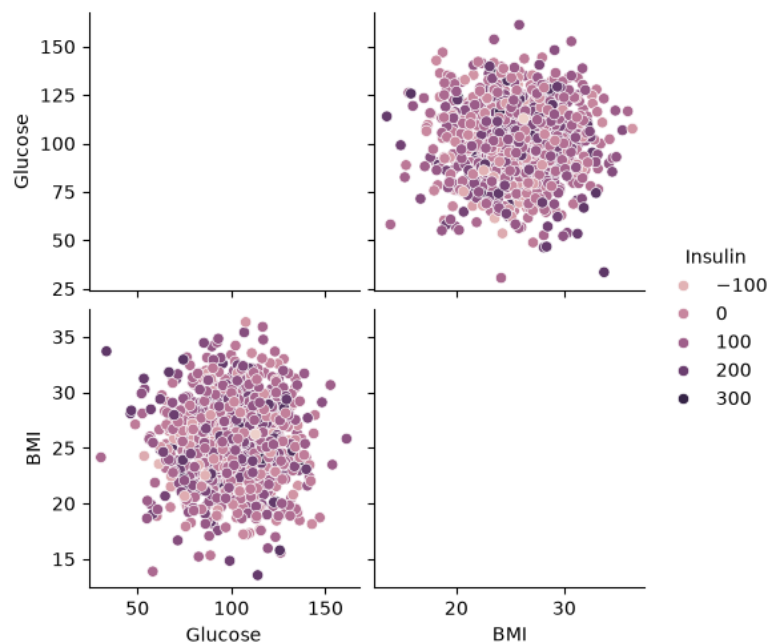


Figure 15: Pairplot of Glucose, BMI, and Insulin indicators.

Inference: Pairwise inspection confirms multivariate separation, illustrating that combined elevation in Glucose and BMI substantially elevates diabetic diagnostic risk.

8 Exploratory Data Analysis 4: Email Spam Classification Dataset

8.1 Dataset Description & Metadata

- **Dataset File:** `spam.csv` (`Mail.ipynb`)
- **Sample Dimensions:** 5,572 raw records (`v1`: target label, `v2`: text payload, and 3 metadata columns: `Unnamed: 2`, `Unnamed: 3`, `Unnamed: 4`).
- **Target Attribute:** `v1` (Ham [legitimate] vs. Spam [unsolicited]).
- **Data Integrity:** Unneeded metadata columns dropped; 403 duplicate message rows removed to prevent evaluation leakage.
- **EDA Execution Call:** `perform_eda("spam.csv")`

8.2 Visual Diagrams & Inferences

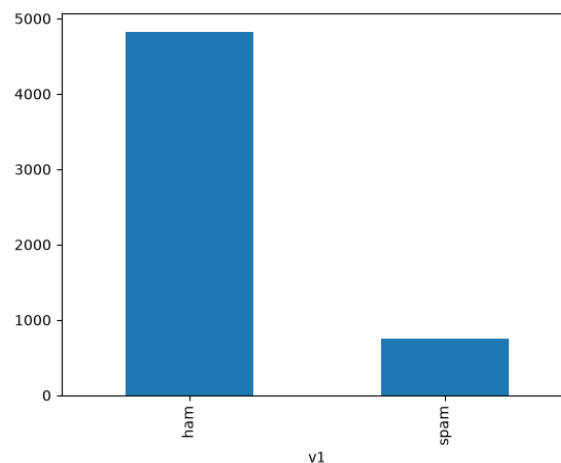


Figure 16: Bar chart depicting class imbalance between Ham and Spam messages.

Inference: The target variable exhibits severe class imbalance (87% *ham* vs. 13% *spam*), confirming that accuracy alone is an insufficient metric and macro F1-score or Precision-Recall metrics are required.

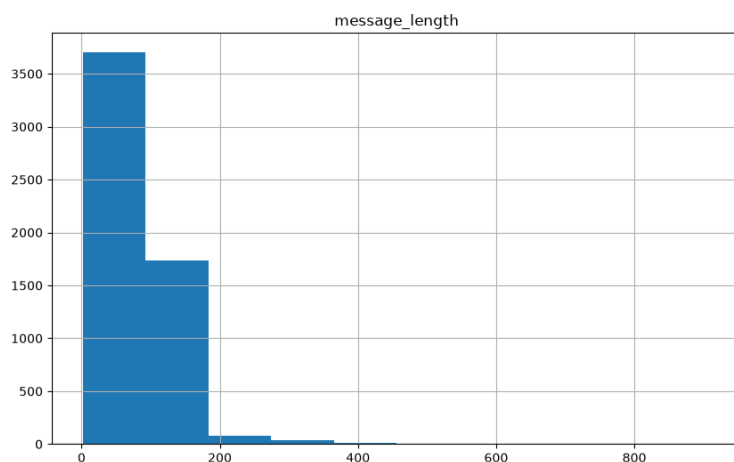


Figure 17: Histograms of extracted message feature attributes.

Inference: Message length and character count distributions follow right-skewed profiles. Spam messages consistently display longer character lengths and higher frequencies of commercial trigger tokens.

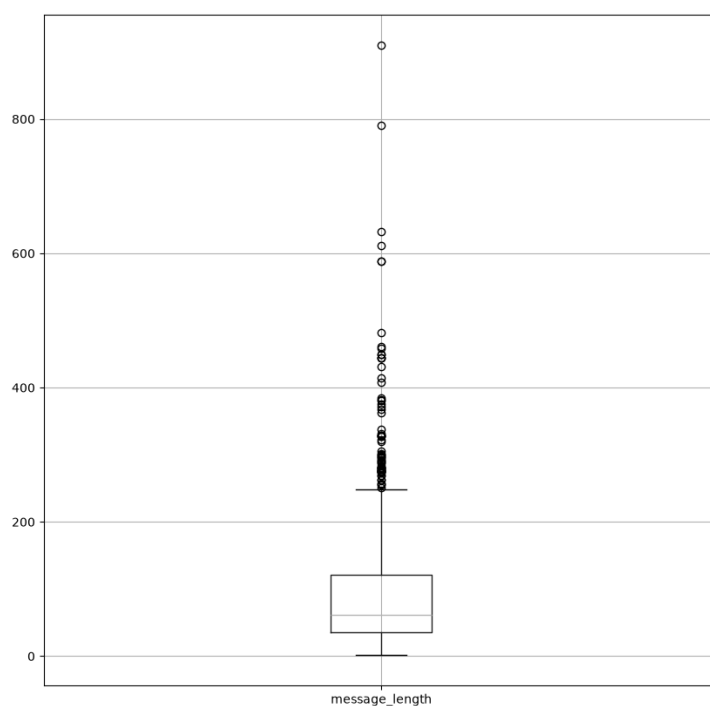


Figure 18: Boxplots showing feature dispersion and upper outliers across text metrics.

Inference: Legitimate ham messages remain tightly bounded within lower text length ranges, whereas spam communications exhibit wide dispersion and high-value upper outliers.

9 Exploratory Data Analysis 5: Handwritten Recognition / MNIST Dataset

9.1 Dataset Description & Metadata

- **Dataset File:** `mnist_test.csv` / `mnist.csv` (`Mnist.ipynb`)
- **Sample Dimensions:** 10,000 samples, 785 columns (1 target label `label` + 784 numerical pixel intensity attributes representing flattened 28×28 grayscale images).
- **Target Attribute:** `label` (10 digit classes: 0 through 9 – balanced 892 to 1,135 samples per class).
- **Data Integrity:** 0 missing values across all 784 pixel columns. Pixel values range between 0 (black background) and 255 (white foreground stroke).
- **EDA Execution Call:** `perform_eda("mnist.csv")`

9.2 Visual Diagrams & Inferences

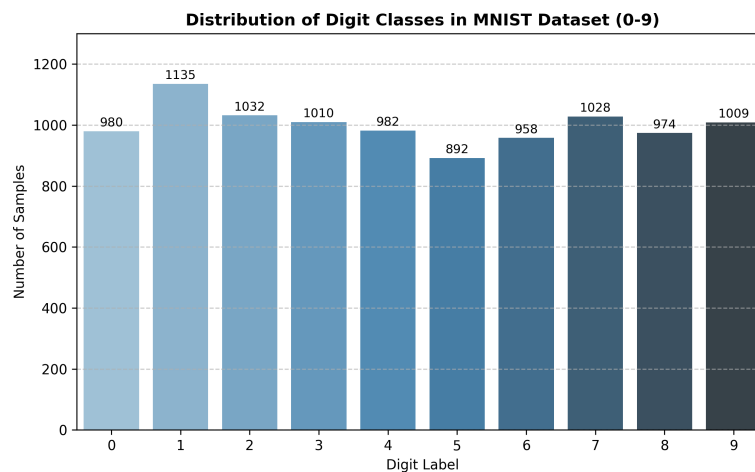


Figure 19: Class frequency distribution across digit labels 0 to 9 in the MNIST dataset.

Inference: Target digit classes are evenly distributed across all 10 categories (ranging from 892 samples for Digit 5 to 1,135 samples for Digit 1), supporting well-balanced multi-class pattern recognition.

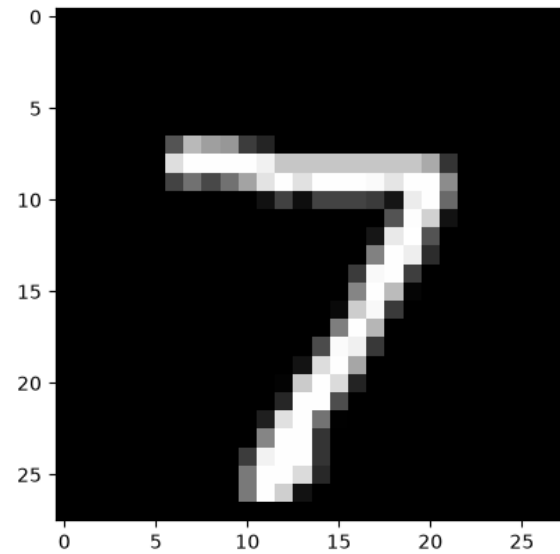


Figure 20: Visual reconstruction of a 28×28 grayscale digit image from flat pixel vector array.

Inference: Reshaping the 784 continuous pixel attributes into a 2D 28×28 matrix reconstructs spatial digit stroke contours, illustrating how raw numerical matrices encode visual pattern geometry.

10 Master Summary Inference Table

Table 2 synthesizes the visual diagrams used, key empirical observations, and analytical inferences across all five benchmark dataset modules.

Dataset	Visual Diagrams Used	Key Empirical Finding	Analytical Takeaway / Inference
Iris Dataset	Histograms, Boxplot, Scatter, Heatmap, Pairplot	Petal length/width exhibit bimodal distribution; $r = 0.96$ between petal metrics.	<i>Iris-setosa</i> is linearly separable along petal dimensions; versicolor/virginica show mild boundary overlap.
Loan Amount Prediction	Histograms, Boxplots, Scatter, Heatmap, Pairplot	$r = 0.93$ between <code>income_annum</code> and <code>loan_amount</code> ; large asset scale variance.	Loan approval thresholds scale linearly with annual earnings; feature scaling is essential for distance-based models.
Predicting Diabetes	Histograms, Boxplot, Scatter, Heatmap, Pairplot	Glucose correlates highest with diabetic diagnosis ($r = 0.47$); 374 zero-insulin values.	Zero values represent biological missing data requiring imputation; elevated Glucose + BMI increases diabetic risk.
Classification of Email Spam	Class Bar Chart, Histograms, Boxplots	Target class imbalance (87% ham vs 13% spam); spam messages display longer text length.	Precision, Recall, and F1-score are necessary evaluation metrics; text length serves as a key predictor.
Handwritten Recognition / MNIST	Class Distribution Bar, 28×28 Grayscale Render	Balanced 10-class distribution (10% per digit); 784 pixel features range $[0, 255]$.	Reshaping flat 784-vector to 28×28 matrix restores spatial digit geometry for computer vision tasks.

Table 2: Master Summary Inference Table synthesizing findings across all 5 benchmark datasets.

11 Preprocessing & Data Hygiene Synthesis Table

Table 3 details the data quality defects identified during EDA and the actionable preprocessing steps required prior to machine learning model construction.

Dataset	Identified Finding	EDA	Required Preprocessing Action
Iris Dataset	Clean data, high scale similarity		Drop non-informative <code>Id</code> column; label-encode categorical target <code>Species</code> .
Loan Amount Prediction	Large numerical scale variance		Strip header whitespace; apply <code>StandardScaler</code> for distance-based algorithms.
Predicting Diabetes	Physiological zero-value entries		Impute zero anomalies in <code>Glucose</code> , <code>BMI</code> , <code>Insulin</code> using median/mean.
Classification of Email Spam	Class imbalance & text payload		Drop metadata columns; remove 403 duplicates; vectorize text payload via TF-IDF.
Handwritten MNIST	High dimensionality (784 features)		Rescale pixel intensities to $[0, 1]$; apply feature selection (<code>SelectKBest</code>).

Table 3: Summary of EDA findings and actionable data preprocessing steps across datasets.

12 Reflection on Learning Outcomes

Through the successful execution of Experiment 1, comprehensive technical and analytical competencies were acquired across Python scientific computing libraries:

- Data Manipulation & Inspection (Pandas & NumPy):** Mastered loading structured datasets across CSV formats, inspecting attribute data types, checking missing values via `.isnull().sum()`, identifying duplicate rows via `.duplicated().sum()`, and performing array reshaping (`.reshape(28, 28)`) for spatial image matrix reconstruction.
- Exploratory Data Visualization (Matplotlib & Seaborn):** Developed expertise in generating publication-quality visual diagnostics:
 - Histograms:* Interpreting Gaussian vs. right-skewed vs. bimodal feature distributions.
 - Boxplots:* Identifying interquartile ranges, feature scale disparities, and extreme upper-tail outliers.
 - Scatter Plots & Pairplots:* Analyzing pairwise relationships, multi-class boundary separability, and cluster formations.
 - Heatmaps:* Quantifying linear Pearson correlation coefficients (r) to identify feature collinearity and primary diagnostic predictors.
- Domain-Specific Preprocessing Awareness:** Gained practical insight into handling real-world data characteristics:
 - Distinguishing between true missing values (`NaN`) and implicit physiological zero-value anomalies (e.g., zero insulin/BMI in clinical data).
 - Recognizing class imbalance in natural language text payloads (e.g., ham vs. spam) and selecting appropriate evaluation strategies.

- Understanding the necessity of feature scaling (**StandardScaler**) when working with features spanning vastly different numerical ranges.